Sheet 2

Preparations and complex ensemble freedom

1. For $\rho = \mathrm{diag}(4/5, 1/5)$, compute purity and the variance of $E = |0\rangle\langle 0|$. Verify that the equally weighted states $\psi_\pm = \sqrt{4/5}\,|0\rangle \pm \sqrt{1/5}\,|1\rangle$ give the same density operator. Are these two states orthogonal?
2. Construct a label-space unitary in the lecture's convention $x_i = \sum_j u_{ij} y_j$, where $x_0 = |0\rangle/\sqrt{2}$, $x_1 = |1\rangle/\sqrt{2}$ and $y_\pm = (|0\rangle \pm i\,|1\rangle)/2$. Check both the relation and $uu^\dagger = \mathbb{1}$. Also construct Problem 1's own label unitary, relating $x_0 = \sqrt{4/5}\,|0\rangle$, $x_1 = \sqrt{1/5}\,|1\rangle$ to $y_\pm = \psi_\pm/\sqrt{2}$. Why can no physical unitary map $|0\rangle, |1\rangle$ to that nonorthogonal pair?
3. A coherent amplitude has mean number $\mu = 3/4$. Write the first three probabilities after averaging its phase uniformly. Compute mean number and vacuum probability. Why would averaging only an overall vector phase leave the original pure density operator unchanged?